INCH-POUND

MIL-P-63311 (AR)
AMENDMENT 4
23 April 1991
SUPERSEDING
AMENDMENT 3
8 April 1988

MILITARY SPECIFICATION

PRIMER ELECTRIC, M83
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63311 (AR), dated 29 June 1979, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete "(ALTERNATE)".

1.1: Delete: "(Alternate)".

2.1: Specifications:

MIL-STD-46215: Change "Igniter, Electronic Primer, M63 and Electric Primers, M80A1 and M83" to "Electric Primers, M80A1, M83 and M120".

PAGE 2

2.1: STANDARDS:

Add: "MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes."

Change: "MIL-STD-286 Propellants: Sampling, Inspection and Testing" to "MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing."

Drawing 8847476: Delete: "(Sheet 2)".

3.1.1 Delete: "(Sheet 2)".

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PAGE 3

* 3.3.1: Delete in its entirety and substitute:

"3.3.1 Ignition element charge composition. The moisture content of the ignition element charge composition after drying shall not exceed 0.02 percent."

3.6: After "3", change "or" to "of".

3.7c: Change "chall" to "shall".

PAGE 4

4.3.1: Drawing 8847476: Delete: "(Sheet 2)".

PAGE 6

Table 1, Examination or Test block: Change: "(Dwg. 8847476 - sheet 2" to "(Dwg. 8847476)".

PAGE 7

4.4.2a: Change "Tast" to "Test".

PAGE 8

* 4.4.2.1, Note 1: Delete in its entirety and substitute:

"Note 1. Three samples minimum shall be selected from each drying oven after completion of drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved or rejected. The selected samples must not be dried more than any other unit which could be selected."

PAGE 10

4.4.2.3, Next Higher Assembly block: Delete: "(sheet 2)".

PAGE 11

* 4.4.2.4, Next Higher Assembly block: Delete: "(sheet 2)".

Major 102: Delete "100%" and substitute "0.40%".

Delete: "105 Retainer fails to withstand application of minimum assembly torque. (See note if lacquer is used). 0.40% 3.2 4.5.8.1".

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4.4.2.5, Drawing Number block: Delete: "(sheet 2)".

4.4.2.5, Critical Defect: Add:

"2 Plug missing 100% 3.1.1 Visual".

PAGE 19

4.4.3.2.1: Change: "Beginning with the first lot produced and continuing until three" to "Beginning with the first lot produced, or a lot following a rejected lot in 4.4.3.22, and continuing until three".

4.4.3.2.2: Add to the end of the paragraph: "If a lot is rejected under these criteria, continue testing lots in accordance with 4.4.3.2.1.".

4.5.1, Line 3: Change: "surrent" to "current".

PAGE 20

* 4.5.2.1: Delete in its entirety and substitute:

"4.5.2.1 Igniter charge composition. The moisture content of the igniter charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour dry time shall be used. The upper insulator from the electric ignition element loading assembly may be removed prior to analysis.

* 4.5.2.1.1: Delete in its entirety.

4.5.2.3: Change: "Method 102.1" to "Method 102.1.3".

4.5.3, Line 2: Delete: "(Sheet 2)".

4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3 and 4.5.6.4: Delete in their entirety and substitute:

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"4.5.6 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium chlorate and gum solids	46.1 ± 2.1%
Lead thiocyanate	35.5 ± 2.0%
Charcoal	20.0 ± 1.5%

4.5.6.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70°C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 + 5°C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)\ (100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.5.6.2 Lead thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.01N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follow:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

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4.5.6.3 Potassium chlorate and gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

PAGE 22

5.1 Delete in its entirety and and substitute:

"5.1 Packaging and marking. Primers shall be packaged and marked in accordance with Dwg. 8847545."

5.1.1 Delete in its entirety.

PAGE 23

5.1.2: Delete in its entirety.

6.1: Change: "M456A1 (HEAT) and M490 (TP)" to "M456A1 (HEAT-T), M456A2 (HEAT-T-MP) and M490 (TP-T)".

6.3: Change: "QAR-I" to "QAT-I".

6.5: Delete: "(Alternate", 2 places.

Line 4: Change: "whic" to "which".

PAGE 24

6.5.1, last line: Change: "Method 102.1" to "Method 102.1.3".

* 6.5.2: Delete in its entirety and substitute:

"6.5.2 Electrical resistance. This test shall be performed in accordance with the applicable defect table. Any primer which fails to comply with the requirement specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3."

6.5.3, last line: Change: "e" to "be".

6.7: Change: "Rock Island, IL 01201" to "Rock Island, IL 61299".

Change: "QAR-E" to "QAT-A".

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Figure 1, Title: Change: "PAINT" to "LACQUER".

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